

DERIVATIVE OF EXPONENTIAL FUNCTION:(1) Let $y = e^x$

$$y + \delta y = e^{x+\delta x}$$

$$\delta y = e^{x+\delta x} - y$$

$$\delta y = e^x \cdot e^{\delta x} - e^x$$

$$\delta y = e^x (e^{\delta x} - 1)$$

$$\frac{\delta y}{\delta x} = e^x \left(\frac{e^{\delta x} - 1}{\delta x} \right)$$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = e^x \left(\lim_{\delta x \rightarrow 0} \frac{e^{\delta x} - 1}{\delta x} \right)$$

$$\frac{dy}{dx} = e^x \times \ln e$$

$$= e^x \times 1$$

$$\boxed{\frac{d}{dx}(e^x) = e^x}$$

(2) Let $y = a^x$

$$y + \delta y = a^{x+\delta x}$$

$$\delta y = a^{x+\delta x} - y$$

$$\delta y = a^x \cdot a^{\delta x} - a^x$$

$$= a^x (a^{\delta x} - 1)$$

$$\frac{\delta y}{\delta x} = a^x \left(\frac{a^{\delta x} - 1}{\delta x} \right)$$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = a^x \left(\lim_{\delta x \rightarrow 0} \frac{a^{\delta x} - 1}{\delta x} \right)$$

$$\because \lim_{\delta x \rightarrow 0} \frac{a^{\delta x} - 1}{\delta x} = \ln a$$

$$\frac{dy}{dx} = a^x \cdot \ln a$$

$$\boxed{\frac{d}{dx}(a^x) = a^x \ln a}$$

DERIVATIVE OF THE LOGARITHMIC FUNCTION:(3) Let $y = \ln x$

$$y + \delta y = \ln(x + \delta x)$$

$$\delta y = \ln(x + \delta x) - y$$

$$\delta y = \ln(x + \delta x) - \ln x$$

$$\delta y = \ln \left(\frac{x + \delta x}{x} \right)$$

$$\because \log m - \log n = \log \frac{m}{n}$$

$$\delta y = \ln \left(1 + \frac{\delta x}{x} \right)$$

$$\frac{\delta y}{\delta x} = \frac{1}{\delta x} \ln \left(1 + \frac{\delta x}{x} \right)$$

$$\frac{\delta y}{\delta x} = \frac{1}{x} \cdot \frac{x}{\delta x} \ln \left(1 + \frac{\delta x}{x} \right)$$

$$\frac{\delta y}{\delta x} = \frac{1}{x} \ln \left(1 + \frac{\delta x}{x} \right)^{\frac{x}{\delta x}}$$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \frac{1}{x} \ln \left[\lim_{\delta x \rightarrow 0} \left(1 + \frac{\delta x}{x} \right)^{\frac{x}{\delta x}} \right]$$

$$\because \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$$

$$\frac{dy}{dx} = \frac{1}{x} \ln e$$

$$\because \ell n e = 1$$

$$\boxed{\frac{d}{dx}(\ln x) = \frac{1}{x}}$$

(4) **Let** $y = \log_a x$

$$y + \delta y = \log_a (x + \delta x)$$

$$\delta y = \log_a (x + \delta x) - y$$

$$\delta y = \log_a (x + \delta x) - \log_a x$$

$$\delta y = \log_a \left(\frac{x + \delta x}{x} \right)$$

$$\frac{\delta y}{\delta x} = \frac{1}{\delta x} \log_a \left(1 + \frac{\delta x}{x} \right)$$

$$\frac{\delta y}{\delta x} = \frac{1}{x} \cdot \frac{x}{\delta x} \log_a \left(1 + \frac{\delta x}{x} \right)$$

$$\frac{\delta y}{\delta x} = \frac{1}{x} \cdot \log_a \left(1 + \frac{\delta x}{x} \right)^{\frac{x}{\delta x}}$$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \frac{1}{x} \log_a \left[\lim_{\delta x \rightarrow 0} \left(1 + \frac{\delta x}{x} \right)^{\frac{x}{\delta x}} \right]$$

$$\frac{dy}{dx} = \frac{1}{x} \log_a e$$

$$\frac{dy}{dx} = \frac{1}{x} \frac{1}{\log_e a} = \frac{1}{x \ln a}$$

$$\boxed{\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}}$$

DERIVATIVE OF HYPERBOLIC FUNCTIONS:

$$(1) \quad \frac{d}{dx}(\sinh x) = \cosh x$$

$$(2) \quad \frac{d}{dx}(\cosh x) = \sinh x$$

$$(3) \quad \frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$$

$$(4) \quad \frac{d}{dx}(\coth x) = -\operatorname{cosech}^2 x$$

$$(5) \quad \frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$$

$$(6) \quad \frac{d}{dx}(\operatorname{cosech} x) = -\operatorname{cosech} x \coth x$$

Proof:

$$(1) \quad \text{Let } y = \sinh x$$

$$y = \frac{e^x - e^{-x}}{2}$$

Differentiate w.r.t 'x'

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{2} \frac{d}{dx}(e^x - e^{-x}) \\ &= \frac{1}{2} \left[\frac{d}{dx}(e^x) - \frac{d}{dx}(e^{-x}) \right] \\ &= \frac{1}{2}(e^x + e^{-x}) \end{aligned}$$

$$\boxed{\frac{d}{dx}(\sinh x) = \cosh x}$$

$$(2) \quad \text{Let } y = \cosh x$$

$$y = \frac{e^x + e^{-x}}{2}$$

Differentiate w.r.t 'x'

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{2} \left[\frac{d}{dx}(e^x + e^{-x}) \right] \\ &= \frac{1}{2} \left[\frac{d}{dx}(e^x) + \frac{d}{dx}(e^{-x}) \right] \\ &= \frac{1}{2}(e^x - e^{-x}) \end{aligned}$$

$$\boxed{\frac{d}{dx}(\cosh x) = \sinh x}$$

(3) Let $y = \tanh x$

$$y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Differentiate w.r.t 'x'

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right)$$

$$= \frac{(e^x + e^{-x}) \frac{d}{dx}(e^x - e^{-x}) - (e^x - e^{-x}) \frac{d}{dx}(e^x + e^{-x})}{(e^x + e^{-x})^2} = \frac{(e^x + e^{-x})(e^x + e^{-x}) - (e^x - e^{-x})(e^x - e^{-x})}{(e^x + e^{-x})^2}$$

$$= \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x + e^{-x})^2}$$

$$= \frac{e^{2x} + e^{-2x} + 2 - (e^{2x} + e^{-2x} - 2)}{(e^x + e^{-x})^2}$$

$$= \frac{e^{2x} + e^{-2x} + 2 - e^{2x} - e^{-2x} + 2}{(e^x + e^{-x})^2}$$

$$= \frac{4}{(e^x + e^{-x})^2}$$

$$= \left(\frac{2}{e^x + e^{-x}} \right)^2$$

$$\boxed{\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x}$$

(4) Let $y = \coth x$

$$y = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

Differentiate w.r.t 'x'

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{e^x + e^{-x}}{e^x - e^{-x}} \right)$$

$$= \frac{(e^x - e^{-x}) \frac{d}{dx}(e^x + e^{-x}) - (e^x + e^{-x}) \frac{d}{dx}(e^x - e^{-x})}{(e^x - e^{-x})^2}$$

$$= \frac{(e^x - e^{-x})(e^x - e^{-x}) - (e^x + e^{-x})(e^x + e^{-x})}{(e^x - e^{-x})^2}$$

$$= \frac{(e^x - e^{-x})^2 - (e^x + e^{-x})^2}{(e^x - e^{-x})^2}$$

$$\begin{aligned}
&= \frac{e^{2x} + e^{-2x} - 2 - (e^{2x} + e^{-2x} + 2)}{(e^x - e^{-x})^2} \\
&= \frac{e^{2x} + e^{-2x} - 2 - e^{2x} - e^{-2x} - 2}{(e^x - e^{-x})^2} \\
&= -\frac{4}{(e^x - e^{-x})^2} \\
&= -\left(\frac{2}{e^x - e^{-x}}\right)^2
\end{aligned}$$

$$\boxed{\frac{d}{dx}(\coth x) = -\operatorname{cosech}^2 x}$$

(5) **Let** $y = \operatorname{sech} x$

$$y = \frac{2}{e^x + e^{-x}}$$

Differentiate w.r.t 'x'

$$\frac{dy}{dx} = 2 \frac{d}{dx} \left(\frac{1}{e^x + e^{-x}} \right)$$

$$\frac{dy}{dx} = 2 \left[\frac{(e^x + e^{-x}) \frac{d}{dx}(1) - 1 \frac{d}{dx}(e^x + e^{-x})}{(e^x + e^{-x})^2} \right]$$

$$\frac{dy}{dx} = 2 \left[\frac{0 - (e^x - e^{-x})}{(e^x + e^{-x})^2} \right]$$

$$= -2 \frac{1}{(e^x + e^{-x})^2} (e^x - e^{-x})$$

$$= -\left(\frac{e^x - e^{-x}}{e^x + e^{-x}}\right) \frac{2}{(e^x + e^{-x})}$$

$$\boxed{\frac{d}{dx}(\operatorname{sech} x) = -\tanh x \operatorname{sech} x}$$

(6) **Let** $y = \operatorname{cosech} x$

$$y = \frac{2}{e^x - e^{-x}}$$

Differentiate w.r.t 'x'

$$\frac{dy}{dx} = 2 \frac{d}{dx} (e^x - e^{-x})^{-1}$$

$$= -2(e^x - e^{-x})^{-2} \frac{d}{dx}(e^x - e^{-x})$$

$$\begin{aligned}
 &= -2 \frac{e^x + e^{-x}}{(e^x - e^{-x})^2} \\
 &= -\frac{e^x + e^{-x}}{e^x - e^{-x}} \cdot \frac{2}{e^x - e^{-x}} \\
 \frac{d}{dx}(\operatorname{cosech} x) &= -\coth x \operatorname{cosech} x
 \end{aligned}$$

Inverse Hyperbolic Functions

$$(1) \quad \sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$$

$$(2) \quad \cosh^{-1} x = \ln(x + \sqrt{x^2 - 1})$$

(1) **Let** $y = \sinh^{-1} x$ for $x, y \in R$ then

$$\sinh y = x$$

$$x = \sinh y$$

$$x = \frac{e^y - e^{-y}}{2}$$

$$2x = e^y - e^{-y}$$

$$2x = e^y - \frac{1}{e^y}$$

$$2x = \frac{e^{2y} - 1}{e^y}$$

$$2e^y x = e^{2y} - 1$$

$$(e^y)^2 - 2x(e^y) - 1 = 0$$

Which is quadratic in e^y . Now we have

$$e^y = \frac{-(-2x) \pm \sqrt{(-2x)^2 - 4(1)(-1)}}{2(1)}$$

$$= \frac{2x \pm \sqrt{4x^2 + 4}}{2}$$

$$= \frac{2x \pm 2\sqrt{x^2 + 1}}{2}$$

$$e^y = x \pm \sqrt{x^2 + 1}$$

As e^y is positive for $y \in R$

$$\text{So } e^y = x + \sqrt{x^2 + 1}$$

$$\ln(e^y) = \ln(x + \sqrt{x^2 + 1})$$

$$y = \ln(x + \sqrt{x^2 + 1})$$

$$\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$$

(2) Let $y = \cosh^{-1} x$ (for $x \in (1, \infty)$, $y \in (0, \infty)$)

$$\cosh y = x$$

$$x = \cosh y$$

$$x = \frac{e^y + e^{-y}}{2}$$

$$2x = e^y + e^{-y}$$

$$2x = e^y + \frac{1}{e^y}$$

$$2x = \frac{e^{2y} + 1}{e^y}$$

$$2xe^y = e^{2y} + 1$$

$$e^{2y} - 2xe^y + 1 = 0$$

$$(e^y)^2 - 2x(e^y) + 1 = 0$$

Which is quadratic in e^y , then we have

$$e^y = \frac{-(-2x) \pm \sqrt{(-2x)^2 - 4(1)(1)}}{2(1)}$$

$$= \frac{2x \pm \sqrt{4x^2 - 4}}{2}$$

$$= \frac{2x \pm 2\sqrt{x^2 - 1}}{2}$$

$$= x \pm \sqrt{x^2 - 1}$$

As e^y is positive

$$\text{So } e^y = x + \sqrt{x^2 - 1}$$

$$\ln(e^y) = \ln(x + \sqrt{x^2 - 1})$$

$$y = \ln(x + \sqrt{x^2 - 1})$$

$$\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1})$$

DERIVATIVE OF INVERSE HYPERBOLIC FUNCTIONS:

(1) $\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$

(2) $\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}$

(3) $\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}$

(4) $\frac{d}{dx}(\coth^{-1} x) = \frac{1}{1-x^2}$

$$(5) \quad \frac{d}{dx}(\operatorname{sech}^{-1} x) = \frac{-1}{x\sqrt{1-x^2}}$$

$$(6) \quad \frac{d}{dx}(\operatorname{cosech}^{-1} x) = \frac{-1}{x\sqrt{1+x^2}}$$

Proof:

(1) **Let** $y = \sinh^{-1} x$
 $\sinh y = x$
 Differentiate w.r.t 'x'

$$\frac{d}{dx}(\sinh y) = \frac{d}{dx}(x)$$

$$\cosh y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\cosh y} \quad (\cosh^2 y = 1 + \sinh^2 y)$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1 + \sinh^2 y}}$$

$$\boxed{\frac{dy}{dx} = \frac{1}{\sqrt{1+x^2}}}$$

(2) **Let** $y = \cosh^{-1} x$
 $\cosh y = x$
 Differentiate w.r.t 'x'

$$\frac{d}{dx}(\cosh y) = \frac{d}{dx}(x)$$

$$\sinh y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\sinh y}$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{\cosh^2 y - 1}} \quad (\sinh^2 y = \cosh^2 y - 1)$$

$$\boxed{\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2 - 1}}}$$

(3) **Let** $y = \tanh^{-1} x$
 $\tanh y = x$
 Differentiate w.r.t 'x'

$$\frac{d}{dx}(\tanh y) = \frac{d}{dx}(x)$$

$$\operatorname{sech}^2 y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\operatorname{sech}^2 y}$$

$$\frac{dy}{dx} = \frac{1}{1 - \tanh^2 y}$$

$$\therefore \operatorname{sech}^2 y = 1 - \tanh^2 y$$

$$\boxed{\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}}$$

- (4) **Let** $y = \coth^{-1} x$
 $\coth y = x$

Differentiate w.r.t 'x'

$$\frac{d}{dx}(\coth y) = \frac{d}{dx}(x)$$

$$-\operatorname{cosech}^2 y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{-\operatorname{cosech}^2 y}$$

$$\frac{dy}{dx} = \frac{-1}{\coth^2 y - 1}$$

$$= \frac{1}{1 - \coth^2 y}$$

$$\therefore \operatorname{cosech}^2 y = \coth^2 y - 1$$

$$\boxed{\frac{d}{dx}(\coth^{-1} x) = \frac{1}{1 - x^2}}$$

- (5) **Let** $y = \operatorname{sech}^{-1} x$
 $\operatorname{sech} y = x$

Differentiate w.r.t 'x'

$$\frac{d}{dx}(\operatorname{sech} y) = \frac{d}{dx}(x)$$

$$-\operatorname{sech} y \tanh y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = -\frac{1}{\operatorname{sech} y \tanh y}$$

$$\therefore \tanh^2 y = 1 - \operatorname{sech}^2 y$$

$$\text{So } \frac{dy}{dx} = -\frac{1}{\operatorname{sech} y \sqrt{1 - \operatorname{sech}^2 y}}$$

$$\boxed{\frac{d}{dx}(\operatorname{sech}^{-1} x) = -\frac{1}{x\sqrt{1 - x^2}}}$$

- (6) **Let** $y = \operatorname{cosech}^{-1} x$
 $\operatorname{cosech} y = x$

Differentiate w.r.t 'x'

$$\frac{d}{dx}(\operatorname{cosech} y) = \frac{d}{dx}(x)$$

$$-\operatorname{cosech} y \coth y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = -\frac{1}{\operatorname{cosech} y \coth y}$$

$$\therefore \coth^2 y = 1 + \operatorname{cosech}^2 y$$

$$\frac{d}{dx}(\operatorname{cosech}^{-1} x) = -\frac{1}{\operatorname{cosech} y \sqrt{1 + \operatorname{cosech}^2 y}}$$

$$\boxed{\frac{d}{dx}(\operatorname{cosech}^{-1} x) = -\frac{1}{x\sqrt{1 + x^2}}}$$

EXERCISE 2.6**Q.1 Find $f'(x)$ if**

(i) $f(x) = e^{\sqrt{x}-1}$

Solution:

$$f(x) = e^{\sqrt{x}-1}$$

Differentiate w.r.t “x”

$$f'(x) = e^{\sqrt{x}-1} \cdot \frac{d}{dx}(\sqrt{x}-1)$$

$$= e^{\sqrt{x}-1} \cdot \frac{1}{2\sqrt{x}}$$

$$\boxed{f'(x) = \frac{e^{\sqrt{x}-1}}{2\sqrt{x}}}$$

(ii) $f(x) = x^3 e^{\frac{1}{x}}$

Solution:

$$f(x) = x^3 e^{\frac{1}{x}}$$

Differentiate w.r.t “x”

$$f'(x) = x^3 \frac{d}{dx}\left(e^{\frac{1}{x}}\right) + e^{\frac{1}{x}} \frac{d}{dx}(x^3)$$

$$= x^3 e^{\frac{1}{x}} \left(\frac{-1}{x^2}\right) + e^{\frac{1}{x}} (3x^2)$$

$$= -xe^{\frac{1}{x}} + 3x^2 e^{\frac{1}{x}}$$

$$\boxed{f'(x) = xe^{\frac{1}{x}} (3x-1)}$$

(iii) $f(x) = e^x (1 + \ln x)$

Solution:

$$f(x) = e^x (1 + \ln x)$$

Differentiate w.r.t “x”

$$f'(x) = e^x \frac{d}{dx}(1 + \ln x) + (1 + \ln x) \frac{d}{dx}(e^x)$$

$$= e^x \left(\frac{1}{x}\right) + (1 + \ln x) e^x$$

$$\boxed{f'(x) = e^x \left[\frac{1}{x} + 1 + \ln x\right]}$$

(iv) $f(x) = \frac{e^x}{e^{-x} + 1}$

Solution:

$$f(x) = \frac{e^x}{e^{-x} + 1}$$

Differentiate w.r.t “x”

$$\begin{aligned} f'(x) &= \frac{(e^{-x} + 1) \frac{d}{dx}(e^x) - (e^x) \frac{d}{dx}(e^{-x} + 1)}{(e^{-x} + 1)^2} \\ &= \frac{(e^{-x} + 1)e^x + (e^x)(e^{-x})}{(e^{-x} + 1)^2} \\ &= \frac{1 + e^x + 1}{(e^{-x} + 1)^2} \end{aligned}$$

$$\boxed{f'(x) = \frac{2 + e^x}{(e^{-x} + 1)^2}}$$

(v) $f(x) = \ln(e^x + e^{-x})$

Solution:

$$f(x) = \ln(e^x + e^{-x})$$

Differentiate w.r.t “x”

$$\begin{aligned} f'(x) &= \frac{1}{e^x + e^{-x}} \frac{d}{dx}(e^x + e^{-x}) \\ &= \frac{1}{e^x + e^{-x}} [e^x - e^{-x}] \end{aligned}$$

$$\boxed{f'(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}}$$

(vi) $f(x) = \frac{e^{ax} - e^{-ax}}{e^{ax} + e^{-ax}}$

Solution:

$$f(x) = \frac{e^{ax} - e^{-ax}}{e^{ax} + e^{-ax}}$$

Differentiate w.r.t “x”

$$f'(x) = \frac{(e^{ax} + e^{-ax}) \frac{d}{dx}(e^{ax} - e^{-ax}) - (e^{ax} - e^{-ax}) \frac{d}{dx}(e^{ax} + e^{-ax})}{(e^{ax} + e^{-ax})^2}$$

$$\begin{aligned}
 f'(x) &= \frac{(e^{ax} + e^{-ax})(ae^{ax} + ae^{-ax}) - (e^{ax} - e^{-ax})(ae^{ax} - ae^{-ax})}{(e^{ax} + e^{-ax})^2} \\
 &= \frac{a(e^{ax} + e^{-ax})^2 - a(e^{ax} - e^{-ax})^2}{(e^{ax} + e^{-ax})^2} \\
 &= \frac{a(e^{2ax} + e^{-2ax} + 2) - a(e^{2ax} + e^{-2ax} - 2)}{(e^{ax} + e^{-ax})^2}
 \end{aligned}$$

$$f'(x) = \frac{4a}{(e^{ax} + e^{-ax})^2}$$

(vii) $f(x) = \sqrt{\ln(e^{2x} + e^{-2x})}$

Solution:

$$f(x) = \sqrt{\ln(e^{2x} + e^{-2x})}$$

Differentiate w.r.t “x”

$$\begin{aligned}
 f'(x) &= \frac{1}{2\sqrt{\ln(e^{2x} + e^{-2x})}} \frac{d}{dx} \ln[e^{2x} + e^{-2x}] \\
 &= \frac{1}{2\sqrt{\ln(e^{2x} + e^{-2x})}} \frac{1}{(e^{2x} + e^{-2x})} [2e^{2x} - 2e^{-2x}] \\
 &= \frac{1}{2\sqrt{\ln(e^{2x} + e^{-2x})}} \frac{1}{(e^{2x} + e^{-2x})} 2[e^{2x} - e^{-2x}]
 \end{aligned}$$

$$f'(x) = \frac{e^{2x} - e^{-2x}}{(e^{2x} + e^{-2x})\sqrt{\ln(e^{2x} + e^{-2x})}}$$

(viii) $f(x) = \ln \sqrt{e^{2x} + e^{-2x}}$

Solution:

$$f(x) = \ln \sqrt{e^{2x} + e^{-2x}}$$

Differentiate w.r.t “x”

$$\begin{aligned}
 f'(x) &= \frac{1}{\sqrt{e^{2x} + e^{-2x}}} \frac{d}{dx} \sqrt{e^{2x} + e^{-2x}} \\
 &= \frac{1}{\sqrt{e^{2x} + e^{-2x}}} \cdot \frac{1}{2\sqrt{e^{2x} + e^{-2x}}} [2e^{2x} - 2e^{-2x}]
 \end{aligned}$$

$$f'(x) = \frac{e^{2x} - e^{-2x}}{e^{2x} + e^{-2x}}$$

$$\boxed{f'(x) = \tanh 2x}$$

Q.2 Find $\frac{dy}{dx}$ if

(i) $y = x^2 \ln \sqrt{x}$

Solution:

$$y = x^2 \ln \sqrt{x}$$

Differentiate w.r.t “x”

$$\frac{dy}{dx} = x^2 \frac{d}{dx}(\ln \sqrt{x}) + \ln \sqrt{x} \frac{d}{dx}(x^2)$$

$$= x^2 \frac{1}{\sqrt{x}} \frac{1}{2\sqrt{x}} + \ln \sqrt{x} \cdot (2x)$$

$$= \frac{x^2}{2x} + 2x \ln \sqrt{x}$$

$$= 2x \ln \sqrt{x} + \frac{x}{2}$$

$$\boxed{\frac{dy}{dx} = x \left(2 \ln \sqrt{x} + \frac{1}{2} \right)}$$

(ii) $y = x\sqrt{\ln x}$

Solution:

$$y = x\sqrt{\ln x}$$

Differentiate w.r.t “x”

$$\frac{dy}{dx} = x \frac{d}{dx} \sqrt{\ln x} + \sqrt{\ln x} \frac{d}{dx}(x)$$

$$= x \frac{1}{2\sqrt{\ln x}} \cdot \frac{1}{x} + \sqrt{\ln x}$$

$$= \frac{x}{2\sqrt{\ln x}} \cdot \frac{1}{x} + \sqrt{\ln x}$$

$$= \frac{1}{2\sqrt{\ln x}} + \sqrt{\ln x}$$

$$\boxed{\frac{dy}{dx} = \frac{1 + 2 \ln x}{2\sqrt{\ln x}}}$$

(iii) $y = \frac{x}{\ln x}$

Solution:

$$y = \frac{x}{\ln x}$$

Differentiate w.r.t “x”

$$\frac{dy}{dx} = \frac{\ln x \frac{d}{dx}(x) - x \frac{d}{dx}(\ln x)}{(\ln x)^2}$$

$$= \frac{\ln x - x \left(\frac{1}{x} \right)}{(\ln x)^2}$$

$$\boxed{\frac{dy}{dx} = \frac{\ln x - 1}{(\ln x)^2}}$$

(iv) $y = x^2 \ln \frac{1}{x}$

Solution:

$$y = x^2 \ln \frac{1}{x}$$

$$= x^2 [\ln 1 - \ln x]$$

$$= x^2 [0 - \ln x]$$

$$y = -x^2 \ln x$$

Differentiate w.r.t “x”

$$\frac{dy}{dx} = - \left[x^2 \cdot \frac{1}{x} + \ln x (2x) \right]$$

$$\boxed{\frac{dy}{dx} = -[x + 2x \ln x]}$$

(v) $y = \ln \sqrt{\frac{x^2 - 1}{x^2 + 1}}$

Solution:

$$y = \ln \sqrt{\frac{x^2 - 1}{x^2 + 1}}$$

$$y = \ln \frac{\sqrt{x^2 - 1}}{\sqrt{x^2 + 1}}$$

$$y = \ln \sqrt{x^2 - 1} - \ln \sqrt{x^2 + 1}$$

$$y = \frac{1}{2} \ln(x^2 - 1) - \frac{1}{2} \ln(x^2 + 1)$$

$$\because \log m^n = n \log m$$

Differentiate w.r.t “x”

$$\frac{dy}{dx} = \frac{1}{2} \frac{d}{dx} \ln(x^2 - 1) - \frac{1}{2} \frac{d}{dx} \ln(x^2 + 1)$$

$$= \frac{1}{2} \cdot \frac{1}{x^2 - 1} (2x) - \frac{1}{2} \frac{(2x)}{x^2 + 1}$$

$$\begin{aligned}
 &= \frac{x}{x^2-1} - \frac{x}{x^2+1} \\
 &= \frac{x(x^2+1) - x(x^2-1)}{(x^2-1)(x^2+1)} \\
 &= \frac{\cancel{x^3} + x - \cancel{x^3} + x}{x^4-1}
 \end{aligned}$$

$$\boxed{\frac{dy}{dx} = \frac{2x}{x^4-1}}$$

(vi) $y = \ln(x + \sqrt{x^2+1})$

Solution:

$$y = \ln(x + \sqrt{x^2+1})$$

Differentiate w.r.t "x"

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{1}{x + \sqrt{x^2+1}} \left[1 + \frac{1(2x)}{2\sqrt{x^2+1}} \right] \\
 &= \frac{1}{x + \sqrt{x^2+1}} \left[\frac{\sqrt{x^2+1} + x}{\sqrt{x^2+1}} \right]
 \end{aligned}$$

$$\boxed{\frac{dy}{dx} = \frac{1}{\sqrt{1+x^2}}}$$

(vii) $y = \ln(9-x^2)$

Solution:

$$y = \ln(9-x^2)$$

Differentiate w.r.t "x"

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{1}{9-x^2} \frac{d}{dx}(9-x^2) \\
 &= \frac{1}{9-x^2} (-2x)
 \end{aligned}$$

$$\boxed{\frac{dy}{dx} = \frac{-2x}{9-x^2}}$$

(viii) $y = e^{-2x} \sin 2x$

Solution:

$$y = e^{-2x} \sin 2x$$

Differentiate w.r.t "x"

$$\begin{aligned}
 \frac{dy}{dx} &= e^{-2x} \frac{d}{dx}(\sin 2x) + \sin 2x \frac{d}{dx}(e^{-2x}) \\
 &= e^{-2x} \cos 2x \cdot 2 + \sin 2x (-2e^{-2x})
 \end{aligned}$$

$$\boxed{\frac{dy}{dx} = 2e^{-2x} [\cos 2x - \sin 2x]}$$

(ix) $y = e^{-x}(x^3 + 2x^2 + 1)$

Solution:

$$y = e^{-x}(x^3 + 2x^2 + 1)$$

Differentiate w.r.t “x”

$$\frac{dy}{dx} = e^{-x} \frac{d}{dx}(x^3 + 2x^2 + 1) + (x^3 + 2x^2 + 1) \frac{d}{dx}(e^{-x})$$

$$\frac{dy}{dx} = e^{-x}[3x^2 + 4x] + (x^3 + 2x^2 + 1)(e^{-x})(-1)$$

$$= e^{-x}(3x^2 + 4x) - e^{-x}(x^3 + 2x^2 + 1)$$

$$= e^{-x}[3x^2 + 4x - x^3 - 2x^2 - 1]$$

$$\boxed{\frac{dy}{dx} = -e^{-x}(x^3 - x^2 - 4x + 1)}$$

(x) $y = xe^{\sin x}$

Solution:

$$y = xe^{\sin x}$$

Differentiate w.r.t “x”

$$\frac{dy}{dx} = x \frac{d}{dx}(e^{\sin x}) + e^{\sin x} \frac{d}{dx}(x)$$

$$= xe^{\sin x} \cdot \cos x + e^{\sin x} \cdot 1$$

$$\boxed{\frac{dy}{dx} = e^{\sin x}(x \cos x + 1)}$$

(xi) $y = 5e^{3x-4}$

Solution:

$$y = 5e^{3x-4}$$

Differentiate w.r.t “x”

$$\frac{dy}{dx} = 5 \frac{d}{dx}(e^{3x-4}) = 5e^{3x-4} \frac{d}{dx}(3x-4)$$

$$= 5e^{3x-4} \cdot (3)$$

$$\boxed{\frac{dy}{dx} = 15e^{3x-4}}$$

(xii) $y = (x+1)^x$

Solution:

$$y = (x+1)^x$$

Taking natural log on both sides,

$$\ln y = \ln(x+1)^x$$

$$\ln y = x \cdot \ln(x+1)$$

Differentiate w.r.t “x”

$$\frac{d}{dx}(\ln y) = x \frac{d}{dx} \ln(x+1) + \ln(x+1) \frac{d}{dx}(x)$$

$$\frac{1}{y} \frac{dy}{dx} = x \cdot \frac{1}{x+1} + \ln(x+1) \cdot 1$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{x}{x+1} + \ln(x+1)$$

$$\frac{dy}{dx} = y \left[\frac{x + (x+1)\ln(x+1)}{x+1} \right]$$

$$\boxed{\frac{dy}{dx} = (x+1)^x \left[\frac{x + (x+1)\ln(x+1)}{x+1} \right]}$$

Or

$$\boxed{\frac{dy}{dx} = (x+1)^x \left[\frac{x}{x+1} + \ln(x+1) \right]}$$

(xiii) $y = (\ln x)^{\ln x}$

Solution:

$$y = (\ln x)^{\ln x}$$

Taking natural log on both sides,

$$\ln y = \ln (\ln x)^{\ln x}$$

$$\ln y = \ln x \cdot \ln (\ln x)$$

Differentiate w.r.t “x”

$$\frac{d}{dx}(\ln y) = \ln x \frac{d}{dx}(\ln(\ln x)) + \ln(\ln x) \frac{d}{dx}(\ln x)$$

$$\frac{1}{y} \frac{dy}{dx} = \cancel{\ln x} \cdot \frac{1}{\cancel{\ln x}} \cdot \frac{1}{x} + \ln(\ln x) \cdot \frac{1}{x}$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{x} + \frac{1}{x} \ln(\ln x)$$

$$\frac{dy}{dx} = y \left[\frac{1 + \ln(\ln x)}{x} \right]$$

$$\boxed{\frac{dy}{dx} = (\ln x)^{\ln x} \left[\frac{1 + \ln(\ln x)}{x} \right]}$$

$$(xiv) \quad y = \frac{\sqrt{x^2-1}(x+1)}{(x^3+1)^{\frac{3}{2}}}$$

Solution:

$$y = \frac{\sqrt{x^2-1}(x+1)}{(x^3+1)^{\frac{3}{2}}}$$

Taking natural log on both sides,

$$\ln y = \ln \frac{(x^2-1)^{\frac{1}{2}}(x+1)}{(x^3+1)^{\frac{3}{2}}}$$

$$\ln y = \ln(x^2-1)^{\frac{1}{2}} + \ln(x+1) - \ln(x^3+1)^{\frac{3}{2}}$$

$$\ln y = \frac{1}{2} \ln(x^2-1) + \ln(x+1) - \frac{3}{2} \ln(x^3+1)$$

Differentiate w.r.t “x”

$$\frac{d}{dx}(\ln y) = \frac{1}{2} \frac{d}{dx}(x^2-1) + \frac{d}{dx} \ln(x+1) - \frac{3}{2} \frac{d}{dx} \ln(x^3+1)$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{2} \frac{1}{x^2-1} (2x) + \frac{1}{x+1} - \frac{3}{2} \cdot \frac{1(3x^2)}{x^3+1}$$

$$= \frac{x}{x^2-1} + \frac{1}{x+1} - \frac{9x^2}{2(x^3+1)}$$

$$= \frac{x}{(x-1)(x+1)} + \frac{1}{x+1} - \frac{9x^2}{2(x+1)(x^2-x+1)}$$

$$= \frac{x(2)(x^2-x+1) + (2)(x-1)(x^2-x+1) - 9x^2(x-1)}{2(x-1)(x+1)(x^2-x+1)}$$

$$= \frac{2x^3 - 2x^2 + 2x + 2(x^3 - 2x^2 + 2x - 1) - 9x^3 + 9x^2}{2(x-1)(x+1)(x^2-x+1)}$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{2x^3 - 2x^2 + 2x + 2x^3 - 4x^2 + 4x - 2 - 9x^3 + 9x^2}{2(x-1)(x+1)(x^2-x+1)}$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{-5x^3 + 3x^2 + 6x - 2}{2(x-1)(x+1)(x^2 - x + 1)}$$

$$\frac{dy}{dx} = \frac{(x-1)^{\frac{1}{2}}(x+1)^{\frac{1}{2}}(\cancel{x+1})}{(x+1)^{\frac{3}{2}}(x^2-x+1)^{\frac{3}{2}}} \left[\frac{-5x^3 + 3x^2 + 6x - 2}{2(x-1)(\cancel{x+1})(x^2-x+1)} \right]$$

$$\frac{dy}{dx} = \frac{(-5x^3 + 3x^2 + 6x - 2)(\cancel{x+1})}{2(x+1)(x^2-x+1)^{\frac{5}{2}}(x-1)^{\frac{1}{2}}(\cancel{x+1})}$$

$$\frac{dy}{dx} = - \frac{5x^3 - 3x^2 - 6x + 2}{2\sqrt{x-1}(x+1)(x^2-x+1)^{\frac{5}{2}}}$$

$$\frac{dy}{dx} = - \frac{5x^2 - 8x + 2}{2\sqrt{x-1}(x^2-x+1)^{\frac{5}{2}}}$$

$$\begin{array}{r} \frac{5x^2 - 8x + 2}{5x^3 - 3x^2 - 6x + 2} \\ \hline \frac{5x^3 \pm 5x^2}{-8x^2 - 6x} \\ \hline \frac{-8x^2 \mp 8x}{2x + 2} \\ \hline \frac{2x + 2}{0} \end{array}$$

Q.3 Find $\frac{dy}{dx}$ if

(i) $y = \cosh 2x$

Solution:

$$y = \cosh 2x$$

Differentiate w.r.t "x"

$$\begin{aligned} \frac{dy}{dx} &= \sinh 2x \frac{d}{dx}(2x) \\ &= \sinh 2x \cdot 2 \end{aligned}$$

$$\boxed{\frac{dy}{dx} = 2 \sinh 2x}$$

(ii) $y = \sinh 3x$

Solution:

$$y = \sinh 3x$$

Differentiate w.r.t "x"

$$\frac{dy}{dx} = \cosh 3x \frac{d}{dx}(3x)$$

$$\boxed{\frac{dy}{dx} = 3 \cosh 3x}$$

(iii) $y = \tanh^{-1}(\sin x)$

Solution:

$$y = \tanh^{-1}(\sin x)$$

Differentiate w.r.t "x"

$$\frac{dy}{dx} = \frac{d}{dx} (\tanh^{-1}(\sin x))$$

$$= \frac{1}{1 - \sin^2 x} \frac{d}{dx} (\sin x)$$

$$= \frac{1}{\cos^2 x} \cos x$$

$$= \frac{1}{\cos x}$$

$$\boxed{\frac{dy}{dx} = \sec x}$$

(iv) $y = \sinh^{-1}(x^3)$

Solution:

$$y = \sinh^{-1}(x^3)$$

Differentiate w.r.t “x”

$$\frac{dy}{dx} = \frac{d}{dx} \sinh^{-1}(x^3)$$

$$= \frac{1}{\sqrt{1 + (x^3)^2}} \frac{d}{dx} (x^3)$$

$$= \frac{1}{\sqrt{1 + x^6}} (3x^2)$$

$$\boxed{\frac{dy}{dx} = \frac{3x^2}{\sqrt{1 + x^6}}}$$

(v) $y = \ln \tanh x$

Solution:

$$y = \ln \tanh x$$

Differentiate w.r.t “x”

$$\frac{dy}{dx} = \frac{1}{\tanh x} \cdot \operatorname{sech}^2 x$$

$$= \coth x \operatorname{sech}^2 x$$

$$= \frac{\cosh x}{\sinh x} \frac{1}{\cosh^2 x}$$

$$= \frac{1}{\sinh x \cosh x}$$

$$= \frac{2}{2 \sinh x \cosh x}$$

$$\boxed{\frac{dy}{dx} = 2 \operatorname{cosech} 2x}$$

(vi) $y = \sinh^{-1}\left(\frac{x}{2}\right)$

Solution:

$$y = \sinh^{-1}\left(\frac{x}{2}\right)$$

Differentiate w.r.t “x”

$$\frac{dy}{dx} = \frac{d}{dx} \left(\sinh^{-1}\left(\frac{x}{2}\right) \right)$$

$$= \frac{1}{\sqrt{1 + \frac{x^2}{4}}} \frac{d}{dx} \left(\frac{x}{2} \right)$$

$$= \frac{1}{\sqrt{4 + x^2}} \frac{1}{2}$$

$$= \frac{1}{\sqrt{4 + x^2}} \frac{1}{2}$$

$$\boxed{\frac{dy}{dx} = \frac{1}{\sqrt{4 + x^2}}}$$